

ORYOL, Russian Federation

WMO: 279060

Lat: 52.9318N

Lon: 36.0189E

Elev: 643

StdP: 14.36

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.2	-4.1	-16.0	2.4	-9.6	-9.8	3.3	-3.8	22.9	26.7	21.5	27.4	5.1	10	0.577

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	86.8	67.9	83.4	66.8	80.5	65.4	70.2	81.5	68.8	79.7	67.3	77.5	7.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
66.5	99.8	75.6	65.0	94.8	73.7	63.6	90.1	72.1	34.5	81.5	33.3	80.0	32.1	77.6	75.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	18.2	16.3	-15.7	90.8	5.7	4.0	-19.8	93.7	-23.1	96.1	-26.3	98.3	-30.4	101.3		
(5)			-16.0	72.8	5.6	1.7	-20.1	74.1	-23.3	75.1	-26.5	76.0	-30.5	77.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	44.1	20.1	20.9	30.3	46.0	57.8	63.9	68.0	65.4	55.3	43.7	32.8	24.0		
	DBStd	19.37	11.55	11.86	8.67	8.48	8.41	6.72	5.97	6.68	7.49	7.94	8.87	11.04		
	HDD50	4139	927	816	611	177	26	2	0	0	32	226	516	806		
	HDD65	7937	1392	1236	1076	571	252	100	36	76	302	660	965	1271		
	CDD50	2000	0	0	0	56	267	419	559	478	190	30	1	0		
	CDD65	326	0	0	0	0	28	68	131	89	10	0	0	0		
	CDH74	2866	0	0	0	10	266	549	1052	883	105	1	0	0		
	CDH80	787	0	0	0	0	46	115	314	301	11	0	0	0		
	WSAvg	7.9	8.9	8.9	9.0	8.4	7.6	7.2	6.4	6.6	6.7	7.8	8.6	8.7		
	PrecAvg	25.2	1.9	1.5	1.5	1.7	1.9	2.7	3.3	2.2	2.5	2.3	1.9	1.7		
(15) Precipitation	PrecMax	33.2	3.7	3.0	3.5	3.6	3.5	5.6	6.7	6.1	7.2	4.2	3.3	5.3		
	PrecMin	17.0	0.9	0.6	0.4	0.1	0.6	0.6	1.0	0.1	0.2	0.3	0.1	0.5		
	PrecStd	3.5	0.7	0.6	0.8	0.9	0.7	1.3	1.4	1.8	1.4	1.2	0.9	1.0		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.1	40.1	57.5	75.3	84.5	87.5	92.3	94.2	81.6	69.1	55.3	45.9		
		MCWB	38.5	37.4	46.5	55.0	65.1	66.7	68.6	67.4	61.6	56.7	51.7	43.9		
	2%	DB	37.1	37.4	49.5	68.8	80.7	83.2	86.7	87.3	77.3	63.2	49.8	42.2		
		MCWB	35.5	35.6	41.0	53.0	62.3	66.3	68.6	67.8	60.9	55.2	47.5	40.8		
	5%	DB	35.0	35.8	44.6	64.7	76.7	80.1	83.4	82.5	72.3	58.5	46.9	37.8		
		MCWB	33.6	34.0	38.6	51.6	61.2	65.1	68.1	66.5	59.5	52.8	45.0	36.5		
	10%	DB	33.4	34.6	41.1	60.5	72.7	76.8	80.0	78.6	67.7	54.6	44.2	35.6		
		MCWB	32.2	33.1	36.8	49.5	59.1	63.9	66.9	64.5	58.0	50.6	42.5	34.4		
	0.4%	WB	38.7	37.8	46.7	57.8	67.3	70.6	72.8	71.7	64.9	59.4	51.9	44.4		
		MCDB	39.3	39.0	55.9	68.8	81.6	81.3	84.4	82.7	75.8	64.8	54.8	45.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.8	35.9	42.3	54.7	64.8	68.6	70.9	69.4	62.8	56.2	48.0	40.7		
		MCDB	36.7	36.9	48.0	64.2	76.4	79.3	82.3	81.8	72.8	61.7	49.4	41.9		
	5%	WB	33.8	34.4	39.4	52.8	62.6	66.8	69.4	67.8	60.9	53.5	45.3	36.8		
		MCDB	34.8	35.4	43.8	62.4	73.4	76.8	79.9	80.4	68.7	57.9	46.6	37.8		
	10%	WB	32.4	33.3	37.1	50.9	60.6	65.1	68.0	65.9	58.9	51.2	42.5	34.6		
		MCDB	33.4	34.4	40.1	59.4	70.5	74.3	77.3	76.5	66.2	54.4	44.1	35.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.9	10.4	11.9	16.6	18.4	17.6	17.5	19.3	16.7	12.4	7.6	7.8		
		MCDBR	6.0	7.1	16.5	22.9	23.0	21.7	22.1	24.8	24.2	19.2	10.4	7.2		
		MCWBR	5.9	6.5	10.0	11.2	9.9	9.4	8.6	9.3	11.0	11.7	8.7	6.9		
		MCDBR	5.8	6.9	14.6	19.9	20.0	19.0	18.8	22.3	20.4	16.8	9.0	7.2		
	5% WB	MCWBR	5.9	6.6	9.7	10.7	9.6	9.1	8.4	9.4	10.6	11.3	8.2	7.1		
(40) Clear-Sky Solar Irradiance	taub		0.296	0.332	0.355	0.430	0.396	0.385	0.419	0.430	0.390	0.357	0.332	0.300		
	taud		2.126	2.043	2.097	2.097	2.297	2.348	2.269	2.223	2.304	2.356	2.249	2.152		
	Ebn at Noon		208	235	259	253	271	275	262	251	247	230	195	181		
	Edh at Noon		23	34	39	45	39	37	40	39	32	25	20	19		
(44) All-Sky Solar Radiation	RadAvg		266	579	1036	1321	1776	1859	1773	1525	987	499	213	161		
	RadStd		50	69	144	119	137	168	139	151	112	82	43	43		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.26	N/A	N/A	N/A	N/A	N/A	N/A	-288	N/A	+76
(47) Regional (7 neighbors)		+1.31	N/A	N/A	N/A	N/A	N/A	-263	-403	+193	+70

Nomenclature: See separate page